

FPGA HW – Memory Hierarchy

The task of this homework is to optimize several kernels for performance by taking advantage of MicroBlaze's features and the memory hierarchy. For each kernel first establish a baseline, then by reconfiguring MicroBlaze configuration, using compiler options, or re-writing the kernel code optimise kernels 1-4 for highest performance.

The input data used to test the kernels should be randomly generated, and must be stored on the external memory.

In your report, compare the performance of the optimized kernels against the baseline implementation. Afterwards discuss which optimization features were most beneficial, how did they improve performance, and why were some kernels easier to optimize than others?

Kernel 1

```
void kernel1 (int *A, int size, int offset) {  
    int i;  
    for(i = 0; i < size-offset; i++)  
        A[i] += A[i+offset] ;  
}
```

Kernel 2

```
void kernel2 (int *A, int size) {  
    int i;  
    for(i = 3; i < size; i++)  
        A[i] = A[i - 1] + A[i - 2] * A[i - 3];  
}
```

Kernel 3

```
void kernel3 (float *h, float *w, int *idx, int size) {  
    for (int i = 0; i < ARRAY_SIZE; ++i)  
        h[idx[i]] = h[idx[i]] + w[i];  
}
```

Kernel 4

```
float kernel4 (float *A, float *B, int size) {  
    float sum = 0;  
    for (int i = 0; i < size; i++) {  
        float diff = A[i] - B[i];
```

Lab #3.4 – FPGA, Kernel optimization

```
        if (diff > 0) sum = (sum + diff);
    }

    return sum;
}
```

Deliverables

FPGA hardware demonstration.

Four page report containing:

1. Block diagram of the hardware platform(s).
2. A description of your implementation methodology. Visuals like flow charts are welcome.
3. The code included as an appendix.

References and appendices are excluded from the report length.

Rubric

The marks for this task are allocated as follows:

1. 50% for demonstrating correct setup of the memory hierarchy, and MicroBlaze configuration.
2. 20% are allocated for the methodology and depth of analysis.
3. 20% are allocated for the discussion of results.
4. 10% for report writing.

This part of the homework contributes 15% towards the final course grade.